## SCOPE

This procedure establishes a standard structure and guideline for all fire department personnel operating at incidents involving confined space rescues.

The procedure outlines responsibilities for first responders, TRT units, Command Officers, and other fire department personnel responding to such incidents.

## PURPOSE

The purpose of this procedure is to establish guidelines for the response of fire department personnel and equipment to confined space rescue incidents. Because confined space rescue operations present a significant danger to fire department personnel, the safe and effective management of these operations require special considerations. This procedure identifies some of the critical issues which must be included in managing these incidents.

## TACTICAL CONSIDERATIONS

OSHA Regulations Standard 29 CFR 1910.146 Permit-Required Confined Spaces regulates entry into confined spaces for general industry and the rescue service and shall be considered the basis for confined space rescue operations. For emergency response, a confined space is defined as:

- A space large enough for personnel to physically enter.
- A space not designed for continuous employee occupancy.
- An area with limited entry and egress.

Confined spaces include caverns, tunnels, pipes, tanks, mine shafts, utility vaults, and any other location where ventilation and access are restricted by the configuration of the space. These factors may also apply to basements and attics. Confined space incidents may involve injured persons or persons asphyxiated or overcome by toxic substances. Pre-incident planning is an important factor in preparing to handle these types of incidents.

Due to the inherent dangers associated with these operations, the Calimesa Fire Department Risk Management Profile shall be applied to all confined space rescue operations and shall be

continuously reassessed throughout the incident. A phased approach to confined space rescue operations which include Arrival, Pre-entry operations, Entry operations, and Termination, can be utilized to mitigate safely and effectively these high-risk / low-frequency events.

Phase I Arrival.

I. ESTABLISH COMMAND

A. The first arriving company officer shall assume Command and begin an immediate size up of the situation while isolating the immediate hazard area and denying entry to all non-rescue personnel.

B. The first arriving US&R unit that is staffed with a TRT Company Officer should be assigned Rescue Group Supervisor. The TRT Company Officer assigned as Rescue Group Supervisor should remain with his crew. Rescue Group Supervisor responsibilities include:

- Assuming technical rescue operations control. (Assigning Rigging Functions. Victim Rescue/Recovery. Support/Supply).
- Identifying hazards and critical factors.
- Developing a rescue plan and back-up plan.
- Communicating with and directing TRT resources assigned to Rescue Group.
- Informing Command of conditions, actions, and needs during all phases of the rescue operation.

C. Designate a Safety Officer. Considerations for Safety Officer include:

- One of the Regional Special Operations Qualified Safety Officers.
- Any experienced TRT Company Officer assigned to the incident.

*A Safety Officer shall be established prior to the implementation of any rescue plan proposed by the Rescue Group Supervisor.*

D. Following the transfer of Command to a Command Officer, a Technical Advisor should be assigned to join the Command Team at their location to assist in managing personnel and resources engaged in the technical aspects of the incident. The Technical Advisor is responsible for ensuring that the rescue plan developed by Rescue Group

Supervisor and communicated to Command is a sound plan in terms of the safety and welfare of both victim(s) and rescuers. Considerations for the Technical Advisor include:

- Any experienced TRT Company Officer assigned to the incident.

*The Technical Advisor position within the Command Team should be filled prior to the implementation of any rescue plan proposed by Rescue Group Supervisor.*

II. SIZE-UP

A. Secure a witness or responsible party to assist in gathering information to determine exactly what happened. If no witnesses are present, Command may have to look for clues on the scene to determine what happened.

B. Assess the immediate and potential hazards to the rescuers.

C. Isolate immediate hazard area, secure the scene, and deny entry for all non-rescue personnel.

D. Establish communications with victim(s) and determine if non-entry retrieval can be made.

E. Assess on-scene capabilities and determine the need for additional resources.

III. SECONDARY ASSESSMENT

A. Secure the entry permit and any other information about the confined space including diagrams showing entry and egress locations.

B. Determine what products may be stored in the confined space and conduct a HazMat assessment.

C. Determine known hazards present in the confined space; atmospheric, mechanical, electrical, etc.

D. Assess the structural stability of the confined space.

**Phase II Pre-entry Operations**

*It must be determined if this will be a RESCUE operation or a RECOVERY operation based on the survivability profile of the victim(s) which include factors such as the location and condition of the victim(s), and elapsed time since the accident occurred.*

Pre-entry operations shall be conducted under the direction of the Rescue Group Supervisor by trained Technical Rescue Technicians.

I. INITIATE FIRE DEPARTMENT CONFINED SPACE RESCUE PERMIT

A. A confined space permit is required if the space has one or more of the following hazards:

| | |
|---|---|
| Atmospheric hazards | Configuration hazard |
| Engulfment Hazard | Any other recognized hazard |

II. MAKE THE GENERAL AREA SAFE

A. Establish a perimeter determined by factors such as atmospheric conditions, wind direction, structural stability, etc.

B. Consider establishing Lobby to control rescue personnel entering the hazard zone.

C. Stop all unnecessary traffic and park all running vehicles downwind.

D. Provide ventilation to the general area if necessary.

## III. MAKE THE RESCUE AREA SAFE

### A. Hazard Assessment / Atmospheric Monitoring

- Determine exactly what hazards and products are present and conduct atmospheric testing for oxygen level, flammability, and toxicity within the confined space. The hazards identified and the results of atmospheric testing will determine the proper level of PPE to be worn by rescuers.
- Atmospheric monitoring shall be done continuously, and readings shall be communicated to Rescue Group Supervisor at least every 5 minutes. Readings must be obtained by personnel with a thorough knowledge of atmospheric monitoring. This function shall be assigned to a Hazardous Materials response unit.
- Implement Lock-Out / Tag-Out procedure if applicable.
- Take appropriate measures to ensure the structural stability of the confined space.
- Any product that is in or flowing into the confined space must be secured and blanked off if possible.

### B. Ventilation

- Rescue Group Supervisor should assign personnel to establish the proper type of mechanical ventilation of the confined space considering the effects that positive and/or negative pressure ventilation will have on the atmosphere.
- Consider positive and negative ventilation together in a push-pull configuration to obtain the greatest effect from ventilation. Consider negative pressure ventilation if there is only one entry point.
- Ventilation personnel shall work closely with air monitoring personnel to ensure safe atmospheric conditions in the confined space as well as the exhaust area and the general working area.

C. Equipment

- Personal Protective Equipment (PPE) shall include helmet, gloves, proper footwear, goggles, turnouts / Nomex or PBI jumpsuit, and a class III harness at a minimum. Additional PPE may be indicated by the hazard and atmospheric assessment.
- Supplied Air Breathing Apparatus (SABA) or Self-Contained Breathing Apparatus (SCBA) shall be utilized by all entry and back-up personnel. SABA is the breathing apparatus of choice. However, if SCBA must be used, personnel shall maintain line of sight and exit the confined space prior to low air alarm activation, following the 75%-25% rule.
- Air monitoring device that monitors oxygen levels, flammability, and toxicity for the entry team.
- Intrinsically safe communication equipment shall be available for entry personnel. If this equipment is not available, entry personnel may use a tagline for communication or a message relay person.
- Intrinsically safe lighting equipment shall be available for entry personnel. If this equipment is not available, entry personnel may use cyalume type lighting sticks.
- A retrieval system with a back-up system shall be prepared and in place. This may include a vertical or horizontal haul system constructed of ropes, pulleys, and other hardware, with a minimum of a 2:1 mechanical advantage.

**Phase III Entry Operations**

*Entry operations shall be conducted under the direction of Rescue Group Supervisor and by trained Technical Rescue Technicians.*

Unit Assignments for Confined Space Rescue

I. MAKE A SAFE ENTRY

The Rescue Group Supervisor shall be responsible for entry operations. The rescue plan will be discussed by the Rescue Group Supervisor, Safety, Command, and the Technical Rescue Technicians. Rescue Group Supervisor and Safety shall ensure that all personnel operating in the

confined space and the area immediately surrounding the confined space are accounted for and wearing appropriate PPE.

The Rescue Group Supervisor shall be responsible for assigning:

1. A crew to perform rigging functions
2. A crew to perform the victim rescue/recovery function
3. A crew to perform support/supply functions

The Rescue Group Supervisor shall be responsible for preparing a Confined Space Rescue documentation sheet.

The rigging crew is responsible for the following actions: rigging, belaying, rope minding, etc.

The rescue/recovery crew is responsible for making entry to locate and remove the victim.

The support/supply crew is responsible for ensuring that both the rigging crew and rescue/recovery crew have all necessary equipment.

Personnel should be assigned to perform air monitoring around the confined space. Typically, this will be done by personnel entering the confined space; however, on large complex incidents this may require the establishment of an Air Monitoring Group. Air monitoring should be performed for the following conditions:

- Oxygen-deficiency: less than 19.5% oxygen
- Oxygen-enrichment: greater than 23.5% oxygen
- Combustible Gases and Vapors (LEL): any atmosphere containing over 10% of lower explosive limit presents an explosion or fire hazard
- Toxicity: Carbon Monoxide over 35 ppm or Hydrogen Sulfide over 10
- Ppm

All atmospheric readings shall be recorded on a confined space rescue worksheet.

All Rescue Group members will be fully briefed on their assignments after the Rescue Group Supervisor has consulted with the IC, a rescue plan has been formulated, and prior to the commencement of rescue operations. If the situation permits, a backup plan should be in place.

The following actions should be taken prior to entry into the confined space

1. Assure lock out, tag out, blank out procedures are complete.
   - All fixed mechanical devices and equipment capable of causing injury shall be placed in a zero mechanical state (ZMS).

- All electrical equipment (excluding lighting) shall be locked out in the open (off) position with a key type of padlock.
- The key shall remain with the person who places the lock on the equipment.
- In cases where lock out is not possible, equipment shall be properly tagged, and physical security provided.

2. Post non-essential personnel at those areas tagged and blanked or blinded.
3. Assure that all personnel who enter the site are equipped with proper PPE and SABA or SCBA.
4. All entry personnel will have their vitals taken and recorded prior to entry if time permits.
5. Assure one (1) backup team for every entry team.
6. Generally, personnel should enter confined spaces in teams of two, however some situations may require that a lone rescuer enter. Constant communication should be maintained with the entry personnel.
7. No team shall enter a space with pagers or other "non-intrinsically safe devices" unless approved prior to entry, based on atmospheric monitoring.
8. Each entry team shall be equipped with the following items,
    - At least one member shall have an intrinsically safe radio
    - Explosion proof lighting, volume, or other explosion proof light.
    - Atmospheric 4 gas monitor
    - Proper protective gear as deemed necessary by the incident commander. At the very least each member shall wear boots, gloves and helmet, and a Pass device.
    - An entry/egress line shall accompany the first entry team and be anchored at their furthest point of penetration.
    - Some form of rapid extrication/retrieval harness for a victim.
    - If the entry team must enter a vertical shaft greater than eight feet, each member shall wear a personal harness and be attached to a fall arresting system upon entering. (Class III)
    - A victim SABA and supply line or SCBA, if applicable.

Once the best method and location for entry has been determined, teams shall begin entry and reconnaissance/rescue/recovery operations in the space. A log shall be maintained noting the entry times of all personnel.

Entry decisions shall be made based on known locations of victims, safety of the opening, atmospheric readings, and ease of recovery points.

If possible, attempt a two-prong attack to reach the victim(s) if their location is known or suspected.

Teams shall be limited to 30 minutes in any space to include entry and egress.

Each team shall be assigned to rehab upon removal from the space until re-hydrated and vital signs are within normal limits.

Once inside the space:

- Assure adequate interior team communications.
- Assure adequate communications with the entry supervisor. IF visual/voice communication is not possible, rescue members should use OATH.

  O-OK (1 pull on rope)

  A-Advance (2 pulls on rope)

  T-Take up slack (3 pulls on rope)

  H-Help (4 pulls on rope)

- Mark, if necessary, with chalk, cylumes or other method, entry, and movement patterns to assure egress.
- Move towards the suspected victim location as a team. (Beware of elevation differences and unstable footing.)

Once the victim has been located, decide:

- Is this a rescue or recovery?
- If rescue, can a SABA or SCBA unit be placed on the victim?
- Can the victim be easily moved towards the opening with current equipment carried by the team?
- Is an additional team needed to make the move?

Communicate your decision/needs to the Incident Commander.

If the victim is to be moved through an opening which represents the only way out of the space, all team members should be stationed to the egress side of the hole/opening in case the victim becomes lodged. Always try to avoid being blocked in by a victim. If this is not possible, assure the following:

- When the move is made, assure it is made quickly and smoothly, leaving the time the space is blocked for egress as minimal as possible.
- Assure that the exterior personnel, as well as interior teams, are aware of the move and a plan is agreed upon prior to blocking the space.
- Assure that all air lines and connections are clear of the victim and his movement path to assure that no airline problems develop because of the victim becoming entangled or pinching off the lines.

**Phase IV Victim Removal**

Once the victim is set for removal, assure the following:

- Attempt to maintain c-spine control, if possible, based on the space and the victim's condition.
- Use removal systems on the exterior that are applicable to the size and weight of the victim.
- Mechanical advantage systems are much preferred over manual hauling.
- Do not use electric winches, etc., to remove victims.
- Decide if the victim is to be removed headfirst or feet first.
- Avoid the use of wristlets on patients with burns to the extremities.

Once the victim is clear from the space, remove all entry team personnel and equipment.

**Phase V Termination**

**A.** Ensure personnel accountability.

B. Remove all tools and equipment used in the rescue/recovery and return to proper apparatus. In cases of a fatality, consider leaving everything in place until the investigative process has been completed.

C. If entry personnel and/or equipment have been contaminated, proper decontamination procedures shall be followed prior to returning to service.

D. Consider a Post Incident Critique (may be more appropriate at a later date).

E. Return to service after turning the scene over to the responsible party and ensuring the scene is secure.

SCOPE

This procedure establishes a standard structure and guideline for all fire department personnel operating at incidents involving trench rescue operations. The procedure outlines responsibilities for first responders, TRT units, Command Officers, and other fire department personnel responding to such incidents. All other Calimesa Fire Department procedures shall apply to trench rescue operations where applicable.

PURPOSE

The purpose of this procedure is to establish guidelines for the response of fire department personnel and equipment to trench rescue incidents. Because trench rescue operations present a significant danger to fire department personnel, the safe and effective management of these operations requires special considerations. *It shall be the policy of the Calimesa Fire Department that NO personnel shall be allowed into an unsafe trench or excavation.* This procedure identifies some of the critical issues which must be included in managing these incidents.

TACTICAL CONSIDERATIONS

***OSHA Regulations Standard 29 CFR 1926 Subpart P*** regulates excavations for general industry and the rescue service and shall be considered the basis for emergency trench rescue operations. For emergency response, an excavation shall be defined by any depression, hole, trench, or earth wall, man-made or natural, of four feet or greater.

Trench collapses generally occur due to unstable soil conditions combined with improper or inadequate shoring. The potential for additional collapse is considered a primary hazard to personnel. Removing soil or debris, adding weight near the edge of an open cut, vibration (such as vehicle movement), rain, or simply the passage of time, may cause additional collapse at any time during the rescue operation.

The Risk to the inherent dangers associated with these operations, the Calimesa Fire Department Risk Management Profile shall be applied to all trench rescue operations and shall be continuously reassessed throughout the incident. A phased approach to trench rescue operations which include Arrival, Pre-entry operations, Entry operations, and Termination, can be utilized to mitigate safely and effectively these high-risk / low-frequency events.

Phase I Arrival

I. ESTABLISH COMMAND

A. First arriving company officer shall assume Command and begin an immediate size up of the situation while ensuring that apparatus remains at least 50 feet from the location of the trench failure. Command shall announce that Level 1 staging should take place at least 150 feet from the trench failure.

B. First arriving US&R unit that is staffed with a TRT Company Officer should be assigned Rescue Group Supervisor. The TRT Company Officer assigned as Rescue Group Supervisor should remain with his crew. Rescue Group responsibilities include:

- Assuming technical rescue operations control.
- Identifying hazards and critical factors.
- Developing a rescue plan and back-up plan.
- Communicating with and directing TRT resources assigned to Rescue Group.
- Informing Command of conditions, actions, and needs during all phases of the rescue operations

C. Designate a Safety Officer. Considerations for Safety Officer include:

  - One of the Regional Special Operations Qualified Safety Officers.
  - Any experienced TRT Company Officer assigned to the incident.

D. Following the transfer of Command to a Command Officer, a Technical Advisor should be assigned to join the Command Team at their location, to assist in managing personnel and resources engaged in the technical aspects of the incident. The Technical Advisor is responsible for ensuring that the rescue plan developed by Rescue Group and communicated to Command is a sound plan in terms of the safety and welfare of both victim(s) and rescuers. Considerations for the Technical Advisor include:

- One of the Regional Special Operations Qualified Safety Officers.
- Any experienced TRT Company Officer assigned to the incident.

***The Technical Advisor position within the Command Team should be filled prior to the implementation of any rescue plan proposed by Rescue Group Supervisor.***

II. Size-Up

A. Secure a witness or responsible party to assist in gathering information to determine exactly what happened. If no witnesses are present, Command may have to look for clues on the scene to determine what happened.

B. Assess the immediate and potential hazards to the rescuers.

C. Isolate immediate hazard area, secure the scene, and deny entry for all non-rescue personnel.

D. Assess on-scene capabilities and determine the need for additional resources.

Phase II Pre-entry Operations

*It must be determined if this will be a RESCUE operation or a RECOVERY operation based on the survivability profile of the victim(s) which include factors such as the location and condition of the victim(s), and elapsed time since the accident occurred.*

I. MAKE THE GENERAL AREA SAFE

A. Establish a hazard zone perimeter 50 feet from the collapse area.

- Keep all non-essential rescue personnel out of the hazard zone.
- Consider establishing Lobby to control rescue personnel entering the hazard zone.
- Remove all non-essential civilian personnel at least 150 feet away from the collapse area.

B. Control traffic movement.

- Shut down roadway.
- Stage apparatus at least 150 feet from the collapse area.
- Re-route all non-essential traffic at least 300 feet from the collapse area.
- Shut down all heavy equipment operating within 300 feet of the collapse area.

II. MAKE THE RESCUE AREA SAFE

*These pre-entry operations shall be conducted under the direction of Rescue Group by trained Technical Rescue Technicians.*

A. Approach the trench from the ends if possible.

B. Look for unidentified hazards such as fissures or an unstable spoil pile.

C. Assess the spoil pile for improper angle of repose and general raveling.

D. Remove any tripping hazards from around the trench.

E. Place ground pads around the lip of the trench.

F. Secure all hazards in the area: utilities, electric, gas, water, etc.

G. De-water the trench if necessary.

H. Monitor the atmosphere in the trench.

I. Ventilate the trench.

Phase III Entry Operations

*Entry operations shall be conducted under the direction of the Rescue Group Supervisor and by trained Technical Rescue Technicians.*

Rescue Group shall be responsible for entry operations. Rescue Group shall ensure that all personnel operating in the hazard zone are accounted for and wearing appropriate PPE.

I. MAKE THE TRENCH SAFE

A. Place ingress and egress ladders into the trench. There should be at least 2 ladders placed into the trench no more than 50 feet apart.

B. Decide on the shoring system to be used (i.e., hydraulic shore, pneumatic shore, timber shore).

C. Create a safe zone in the non-collapsed area of the trench, from both ends, if possible, by implementing an approved shoring system.